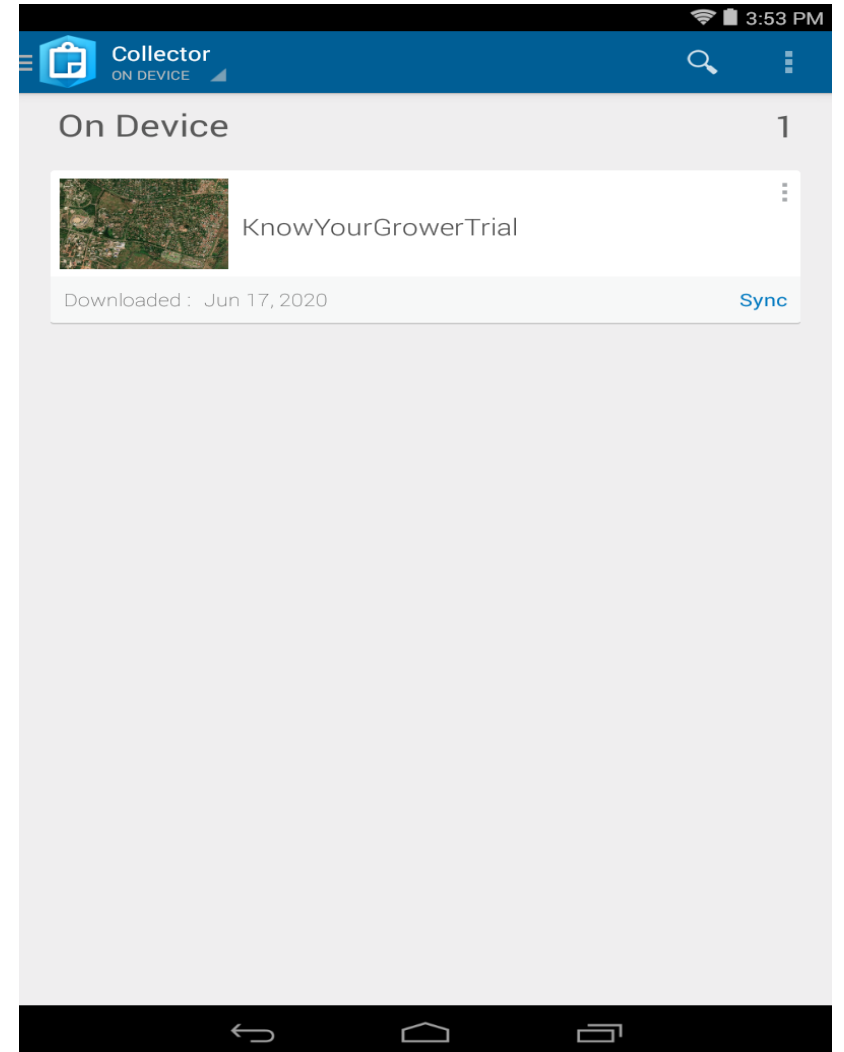


Lecture 3: Using ArcGIS Collector in the Field

- Get to the farm
- Use collector to fill the questionnaire and map the parcel

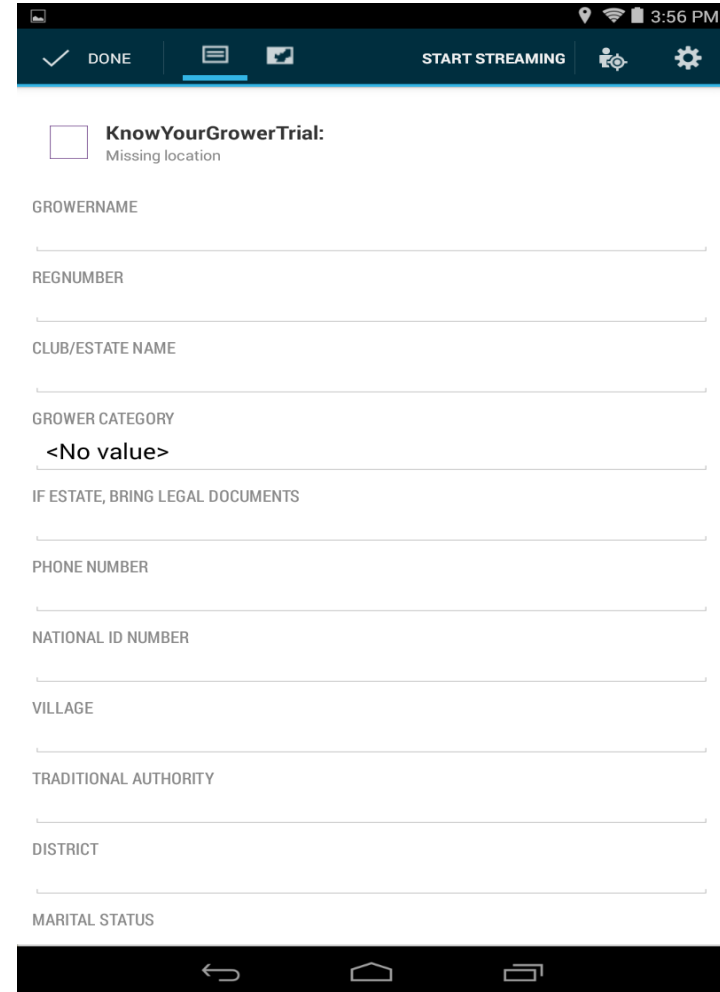
Installation and start mapping

1. Install and start the Collector. This is done as any other app on the tablet.
2. You are already signed in, if not ask the data manager to sign in for you. Note that to sign in you need access to the internet..
3. Select Maps/TrainingTC then **On Device to view offline maps.**
4. Select the appropriate district, TA or enumeration area (EA).
5. The locator will also show the TA or EA in which you will be mapping.



Filling the forms/ questionnaire

- Select the + sign to start mapping.
- Fill the forms for each of the variables.
- Make sure to follow consistent punctuations and capitalization

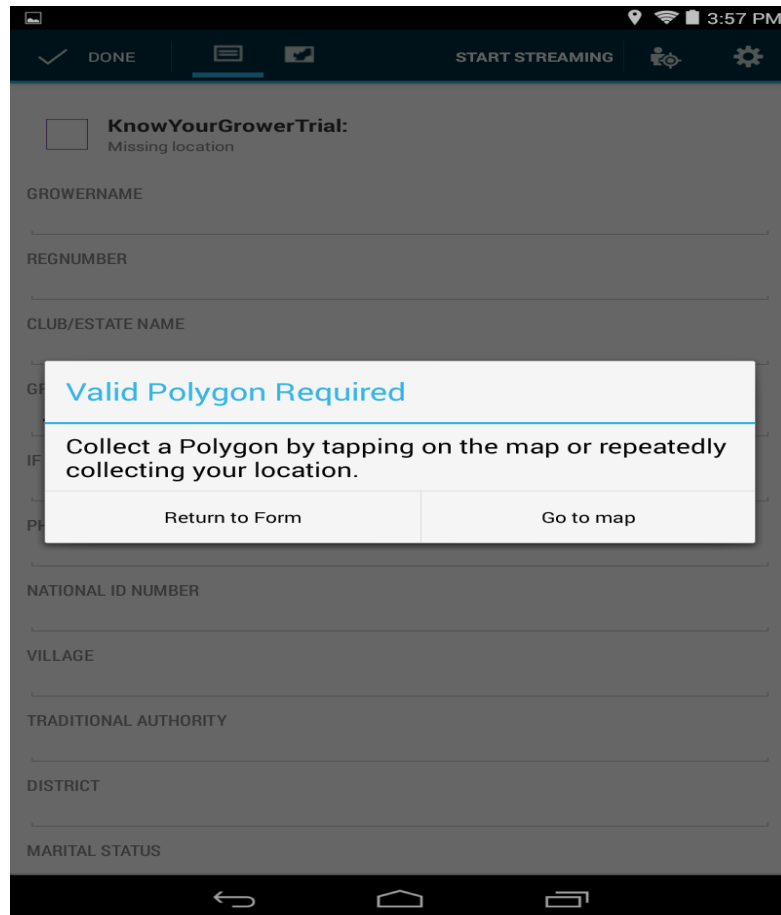


The screenshot shows a mobile application interface for a form titled "KnowYourGrowerTrial:". The form is displayed on a smartphone screen with a status bar at the top showing the time as 3:56 PM. The form fields are as follows:

- KnowYourGrowerTrial:** Missing location (with a location pin icon)
- GROWERNAME** (text input field)
- REGNUMBER** (text input field)
- CLUB/ESTATE NAME** (text input field)
- GROWER CATEGORY** (text input field with the value "<No value>")
- IF ESTATE, BRING LEGAL DOCUMENTS** (text input field)
- PHONE NUMBER** (text input field)
- NATIONAL ID NUMBER** (text input field)
- VILLAGE** (text input field)
- TRADITIONAL AUTHORITY** (text input field)
- DISTRICT** (text input field)
- MARITAL STATUS** (text input field)

The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps icons.

Click done and go to map

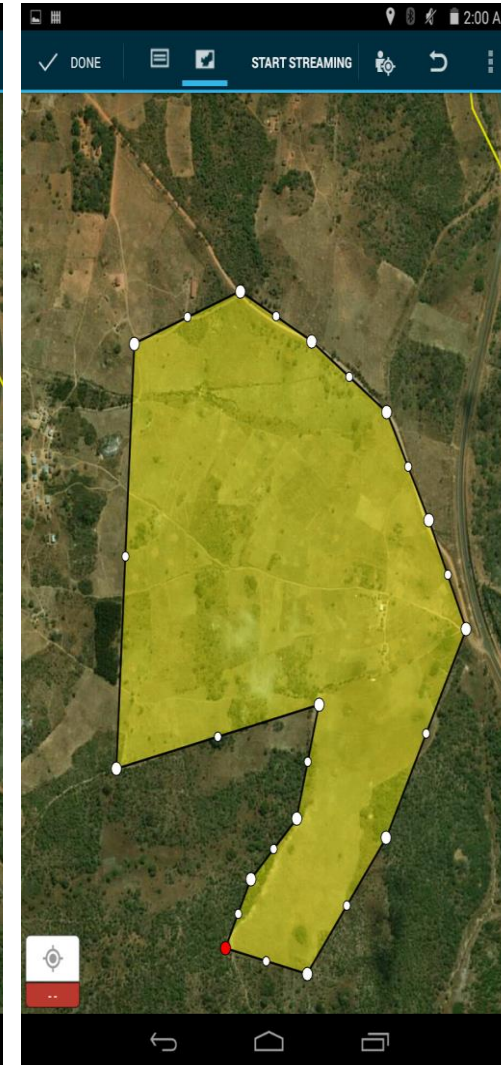


The screenshot shows a mobile application interface with a dark blue header bar. The header contains a checkmark icon, the word 'DONE', a list icon, a map icon, the text 'START STREAMING', a location pin icon, and a settings gear icon. The status bar at the top shows the time as 3:57 PM. The main form area is grey and contains several input fields with labels: 'KnowYourGrowerTrial: Missing location' (with a checkbox), 'GROWERNAME', 'REGNUMBER', 'CLUB/ESTATE NAME', 'GROWER NAME', 'IF', 'P', 'NATIONAL ID NUMBER', 'VILLAGE', 'TRADITIONAL AUTHORITY', 'DISTRICT', and 'MARITAL STATUS'. A white dialog box with a blue border is overlaid on the form. The dialog has a title 'Valid Polygon Required' in blue text, followed by the instruction 'Collect a Polygon by tapping on the map or repeatedly collecting your location.' in black text. At the bottom of the dialog are two buttons: 'Return to Form' and 'Go to map'. The bottom of the screen shows a black navigation bar with a back arrow, a home icon, and a recent apps icon.



Mapping: Method 1

- Zoom in as much as possible until you start seeing recognizable features like roads, forests, houses and farm parcels.
- Click on the positioning tool and a dot should appear on your map showing you where you are. Wait until the accuracy is below 10m.
- Select the + icon and start taking points on the map
- This part should be extremely participatory and the enumerator should walk with the farmer on at least three corners that correspond to the points on the map.



Examples

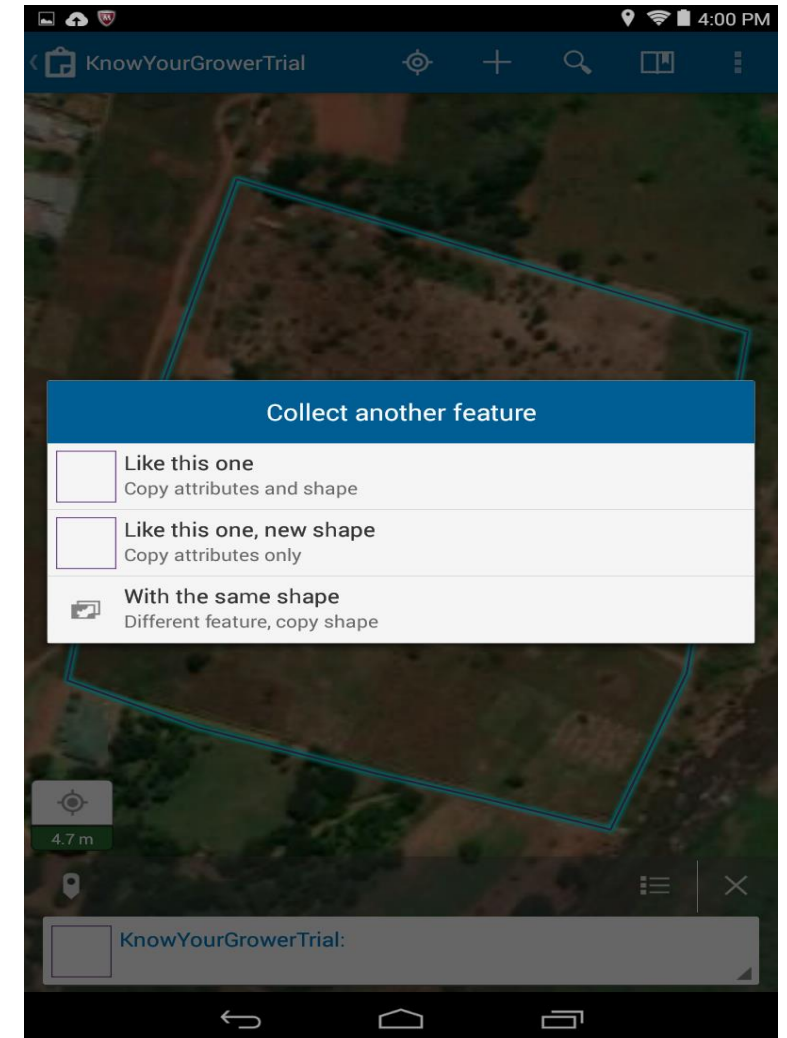
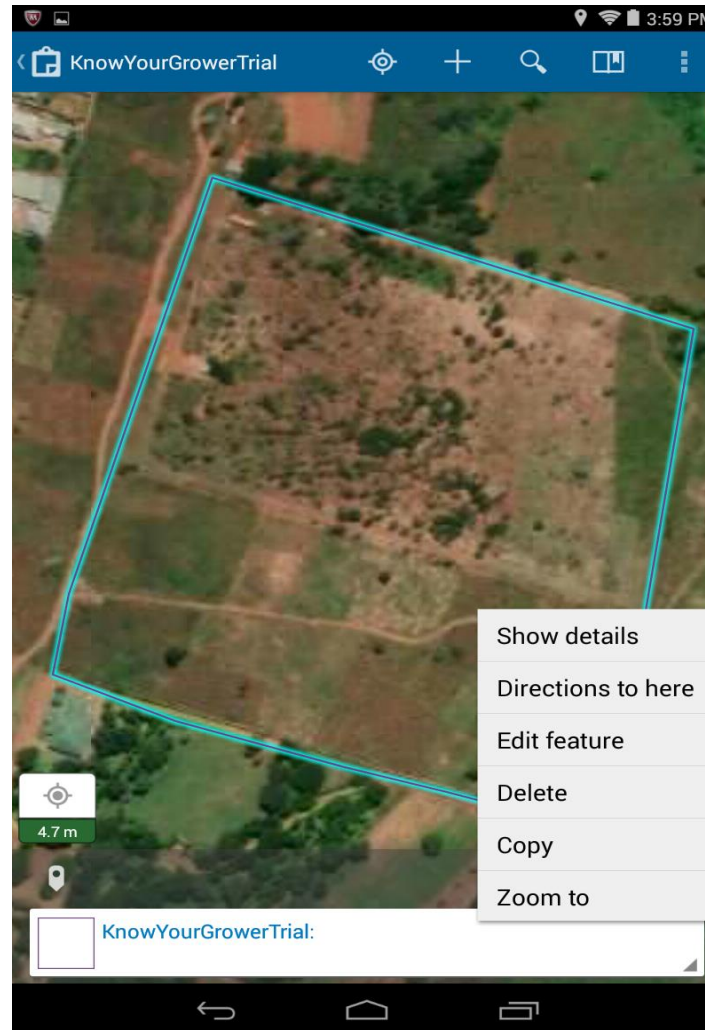


Mapping: Method 2

- If the parcel is small or the farmer can't recognize the features, use this method
- Click start streaming at one corner on the boundary of the estate. Make sure you mark the starting point. Then walk around the estate and click done after finishing the walk.

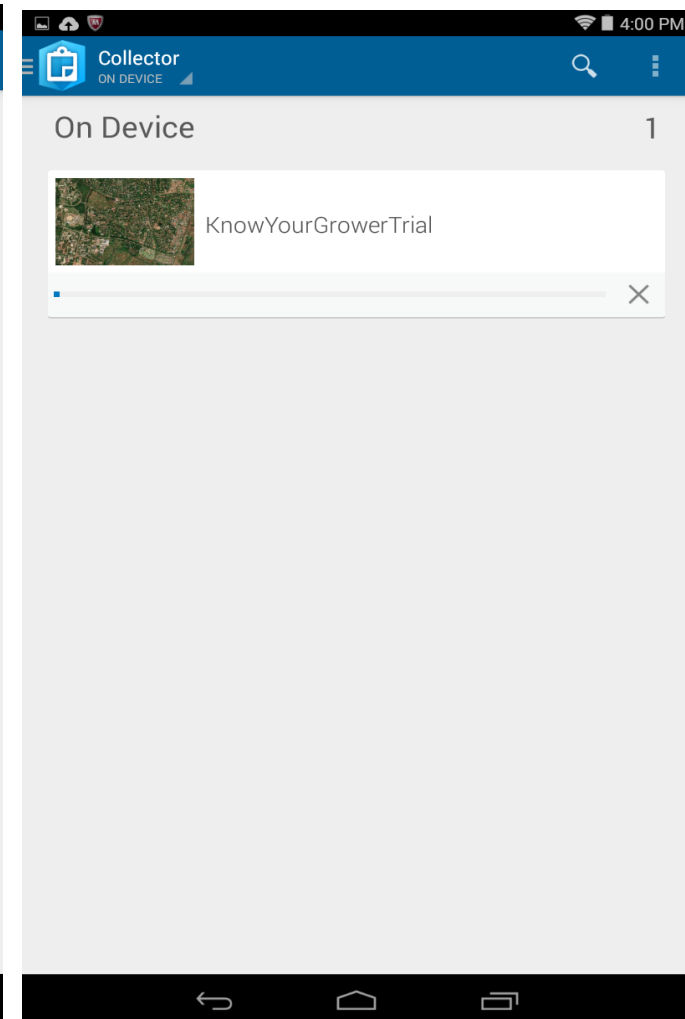
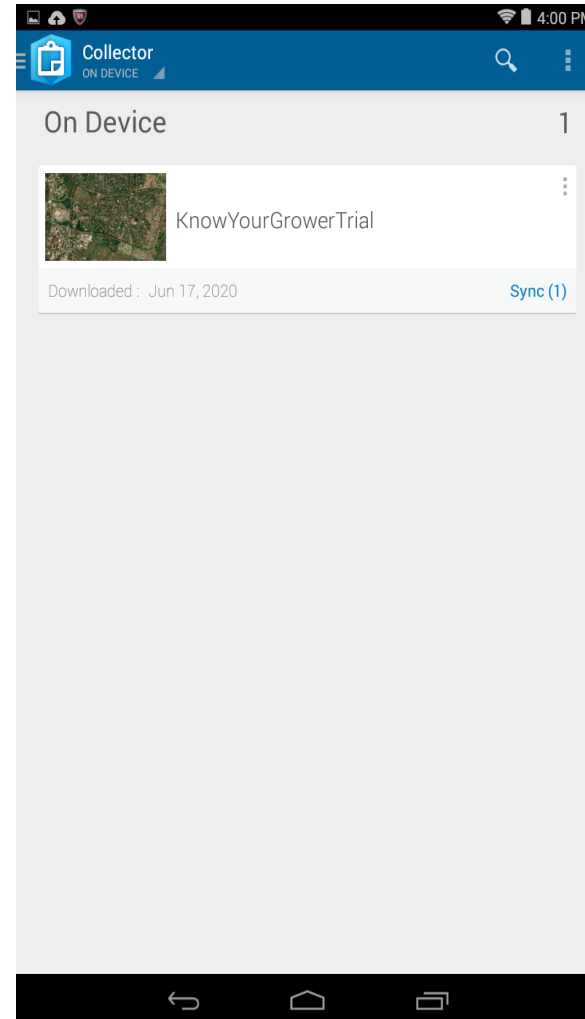
Editing and deleting maps

- To edit, touch the shape you drew- it should then appear as highlighted.
- Click on the lower right corner of the name.
- A pop up will appear with the following options: show details, directions to here, edit feature, delete, copy and zoom to.



Syncing and errors

- As you add the maps to the base map in the collector, you will see the number of maps to be synced to the online server. Click on sync. This does not require a lot of bandwidth and is very quick.



Other tips

- **Direction to village or farm:** The supervisor together with community leaders can identify the crude location of the farm or village from the imagery. The enumerator can then insert small polygon at those locations and use a direction tool on the collector to move from one location to the target location. Note: this polygon should be deleted after use.
- **Measure:** If the enumerator is not sure of a certain distance or area, he/she can use the measure tool to check the distance and area of the farm without recording the information to the arc collector. This can be done quickly to probe the farmer on the shape and area to be measured.
- **Accuracy:** The collector is fixed at 10m accuracy which means that the enumerator should wait until the accuracy is below 10m to start mapping the boundaries of the estate.
- **Open space:** To get a better accuracy, the enumerator should stand or be in an open space clear from tree cover.
- **Collector not working?** If the collector doesn't open or is not loading quick enough, restart the tablet. Makes sure the locations you recently collected are already saved before restarting.
- **If you can't see clearly the screen,** you need to adjust the brightness of the tablet. Go to settings, on brightness select the outdoor option.

Data quality control issues

- Should be before, during and after data collection
- For the enumerator
 - Make sure the farmer participates as much as possible
 - Verify at least two corners of the map drawn
- For the supervisor
 - Check the concordance between the map drawn and the parcels on the farm
 - Check the concordance in the location the map is drawn.
 - Cross check with survey solutions identifiers.
 - Access ArcGIS online to view of the maps. Check for overlaps, and mistakes
 - Let enumerators make the changes or change appropriately
 - Copy files from one master file into other tablets
- For the data manager
 - Check time, exact location and shapes of the polygons. Compare to high resolution image.
 - Do quick analysis of the data and update other stakeholders

More tips

- **Downloading base maps in Arc Collector**
- *Downloading the imagery:* Make sure you have good internet and that you can see all the layers on the map. When choosing the map detail, consider the size of the base map and the ability to map the farm parcels.
- *Level of detail to download:* Previous work, I aimed at getting a detail as low as 1:3000 which is about 18 of 20 levels. The challenge though was that some of the TAs were too big as such compromises were made of up to 1:6000.

More tips

- Syncing when files are transferred across tablets with different usernames
- **Step 1:** Offline- Connect the tablet to PC, change the name of the folder with username to the name of the publisher. Do not change the long text after the username.
- **Step 2:** Online- In Arc Collector, click on the three points on the top right. Click switch account.
- **Step 3:** Sync the maps and switch back to the username for the tablet. Make sure you change the name of the folder back to the original username

More tips

- Copying downloaded basemaps from one tablet to another
- In a heavy field work it is difficult to download imagery with the same level of resolution across say 20 tablets.
- Download all imagery in one tablet
- Connect the tablet to a PC
- Copy the extracted folder in the tablet and paste it in the PC.
- To transfer this folder into another tablet, change the name on the name in the in the offline folder to the username of the folder you are transferring into.